In [ ]:

```
#Name: Meenakshi Sangeeth
#Registration No: 240905630
#Section: PG
#Roll No: 34
```

In [116]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [115]:

```
df=pd.read_csv('deliveries.csv')
df
```

Out[115]:

	match_id	inning	batting_team	bowling_team	over	ball	batter	bowler	non_st
0	335982	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	1	SC Ganguly	P Kumar	McC
1	335982	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	2	BB McCullum	P Kumar	SC Ga
2	335982	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	3	BB McCullum	P Kumar	SC Ga
3	335982	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	4	BB McCullum	P Kumar	SC Ga
4	335982	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	5	BB McCullum	P Kumar	SC Ga
...	...	...	...	...	...	...	...	...	
260915	1426312	2	Kolkata Knight Riders	Sunrisers Hyderabad	9	5	SS Iyer	AK Markram	VF
260916	1426312	2	Kolkata Knight Riders	Sunrisers Hyderabad	9	6	VR Iyer	AK Markram	SS
260917	1426312	2	Kolkata Knight Riders	Sunrisers Hyderabad	10	1	VR Iyer	Shahbaz Ahmed	SS
260918	1426312	2	Kolkata Knight Riders	Sunrisers Hyderabad	10	2	SS Iyer	Shahbaz Ahmed	VF
260919	1426312	2	Kolkata Knight Riders	Sunrisers Hyderabad	10	3	VR Iyer	Shahbaz Ahmed	SS

260920 rows × 17 columns

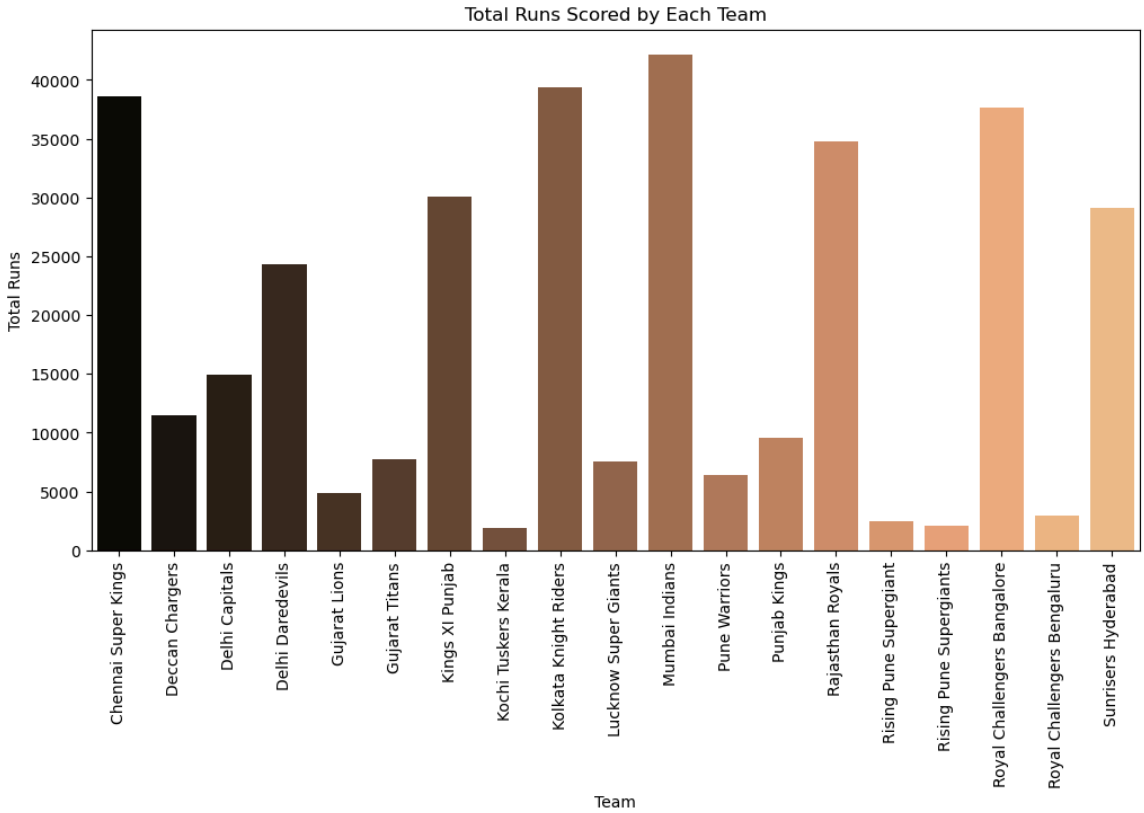
In [44]:

```
#1. Find and visualize the total runs scored by each team


total_runs = df.groupby('batting_team')['total_runs'].sum().reset_index
print(total_runs)
plt.figure(figsize=(12, 6))
sns.barplot(data=total_runs, x='batting_team', y='total_runs', palette=
plt.title('Total Runs Scored by Each Team')
plt.xticks(rotation=90)
plt.xlabel('Team')
plt.ylabel('Total Runs')
plt
```

```
                   batting_team  total_runs
0           Chennai Super Kings       38629
1               Deccan Chargers       11463
2                Delhi Capitals       14900
3             Delhi Daredevils       24296
4                Gujarat Lions        4862
5                Gujarat Titans        7757
6               Kings XI Punjab       30064
7          Kochi Tuskers Kerala        1901
8         Kolkata Knight Riders       39331
9          Lucknow Super Giants        7510
10               Mumbai Indians       42176
11                Pune Warriors        6358
12                Punjab Kings        9536
13             Rajasthan Royals       34747
14       Rising Pune Supergiant        2470
15      Rising Pune Supergiants        2063
16  Royal Challengers Bangalore       37692
17  Royal Challengers Bengaluru        2930
18          Sunrisers Hyderabad       29071
```

Out[44]:

```
<module 'matplotlib.pyplot' from 'C:\\ProgramData\\Anaconda3\\lib\\site-pa
ckages\\matplotlib\\pyplot.py'>
```



In [8]:

```
#2 Idenitfy and display the names of the top 10 highest scoring batters

top_batters = df.groupby("batter")["batsman_runs"].sum().nlargest(10)
print(top_batters)
```

```
batter
V Kohli           8014
S Dhawan          6769
RG Sharma         6630
DA Warner         6567
SK Raina          5536
MS Dhoni          5243
AB de Villiers    5181
CH Gayle          4997
RV Uthappa        4954
KD Karthik        4843
Name: batsman_runs, dtype: int64
```

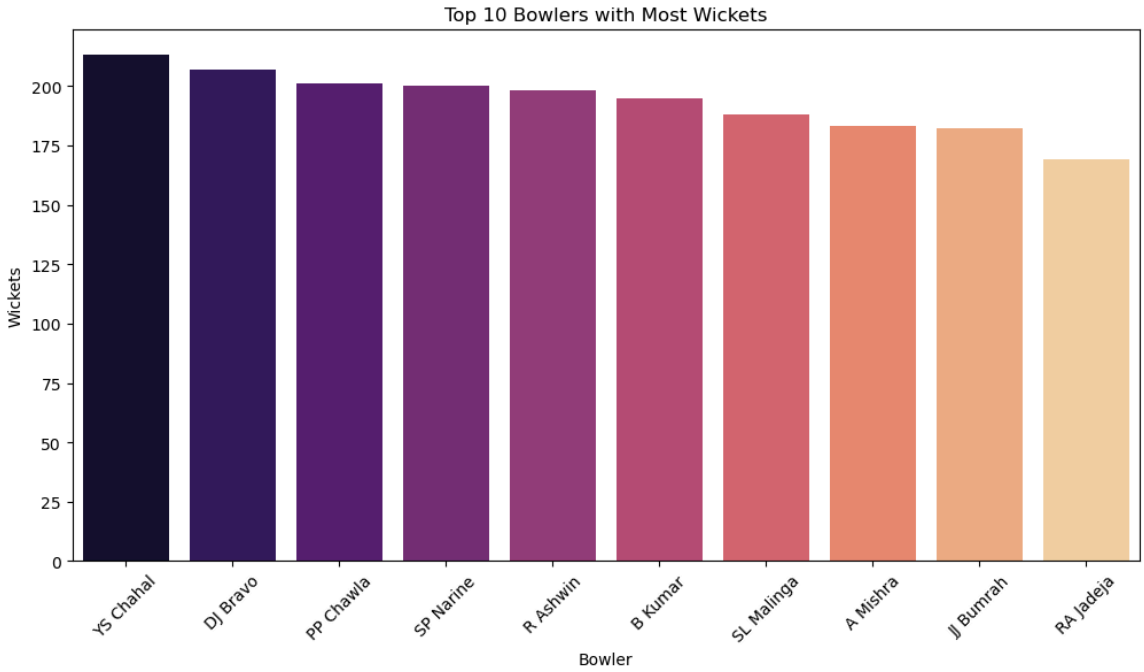
In [117]:

```
#3 Find and visualize the top 10 bowlers with the most wickets

df["is_wicket"] = df["is_wicket"].astype(bool)
top_bowlers = df[df["is_wicket"]].groupby("bowler")["is_wicket"].sum().
print(top_bowlers)
plt.figure(figsize=(12,6))
sns.barplot(x=top_bowlers.index, y=top_bowlers.values, palette="magma")
plt.title("Top 10 Bowlers with Most Wickets")
plt.xlabel("Bowler")
plt.ylabel("Wickets")
plt.xticks(rotation=45)
plt
```

```
bowler
YS Chahal     213
DJ Bravo      207
PP Chawla     201
SP Narine     200
R Ashwin      198
B Kumar       195
SL Malinga    188
A Mishra      183
JJ Bumrah     182
RA Jadeja     169
Name: is_wicket, dtype: int64
```

Out[117]:

```
<module 'matplotlib.pyplot' from 'C:\\ProgramData\\Anaconda3\\lib\\site-pa
ckages\\matplotlib\\pyplot.py'>
```



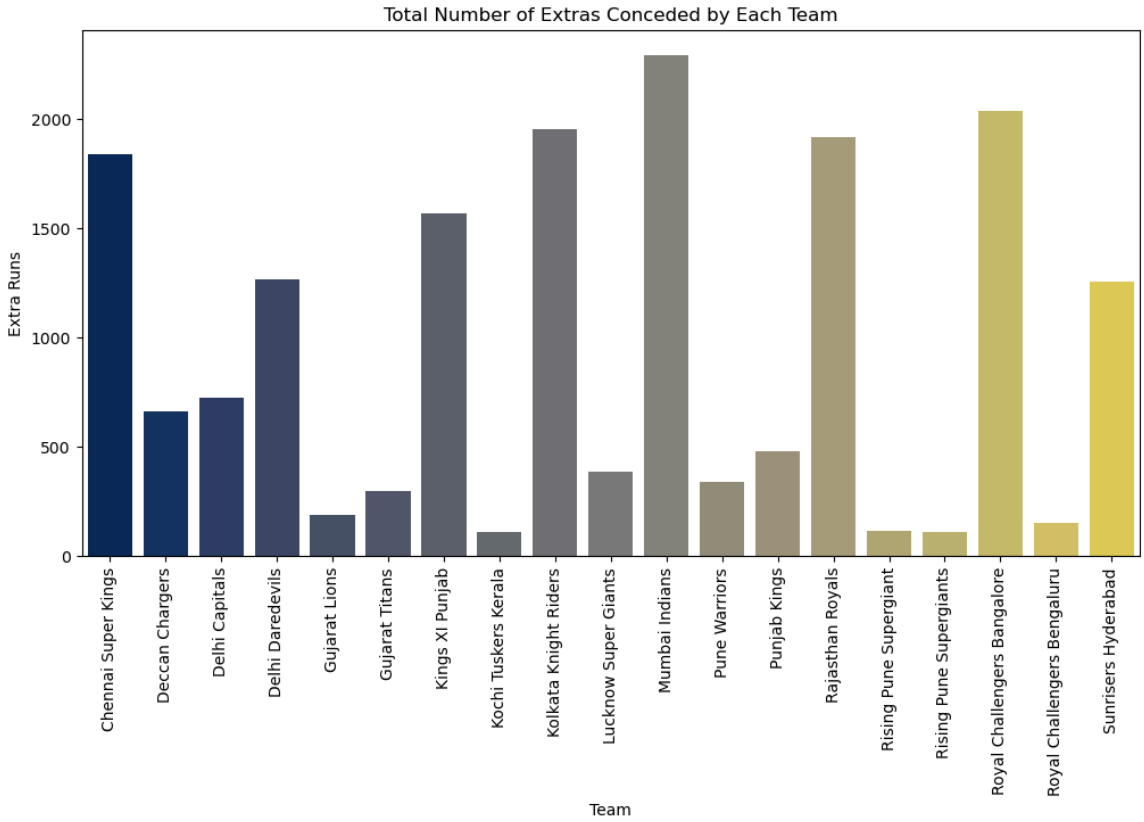
In [52]:

```
#4. Tabulate and visualize the total number of extras conceded by each t

total_extras_team = df.groupby("bowling_team")["extra_runs"].sum().rese
print(total_extras_team)
plt.figure(figsize=(12,6))
sns.barplot(data=total_extras_team, x='bowling_team', y='extra_runs', pa
plt.title("Total Number of Extras Conceded by Each Team")
plt.xlabel("Team")
plt.ylabel("Extra Runs")
plt.xticks(rotation=90)
plt
```

```
                   bowling_team  extra_runs
0           Chennai Super Kings        1842
1               Deccan Chargers         659
2                Delhi Capitals         725
3              Delhi Daredevils        1268
4                 Gujarat Lions         188
5                Gujarat Titans         295
6               Kings XI Punjab        1571
7          Kochi Tuskers Kerala         110
8         Kolkata Knight Riders        1957
9          Lucknow Super Giants         386
10               Mumbai Indians        2295
11                Pune Warriors         335
12                 Punjab Kings         480
13             Rajasthan Royals        1917
14       Rising Pune Supergiant         111
15      Rising Pune Supergiants         108
16  Royal Challengers Bangalore        2040
17  Royal Challengers Bengaluru         151
18          Sunrisers Hyderabad        1254
```

Out[52]:

```
<module 'matplotlib.pyplot' from 'C:\\ProgramData\\Anaconda3\\lib\\site-pa
ckages\\matplotlib\\pyplot.py'>
```


Total Number of Extras Conceded by Each Team
Extra Runs
0
500
1000
1500
2000
Chennai Super Kings
Deccan Chargers
Delhi Capitals
Delhi Daredevils
Gujarat Lions
Gujarat Titans
Kings XI Punjab
Kochi Tuskers Kerala
Kolkata Knight Riders
Lucknow Super Giants
Mumbai Indians
Pune Warriors
Punjab Kings
Rajasthan Royals
Rising Pune Supergiant
Rising Pune Supergiants
Royal Challengers Bangalore
Royal Challengers Bengaluru
Sunrisers Hyderabad
Team

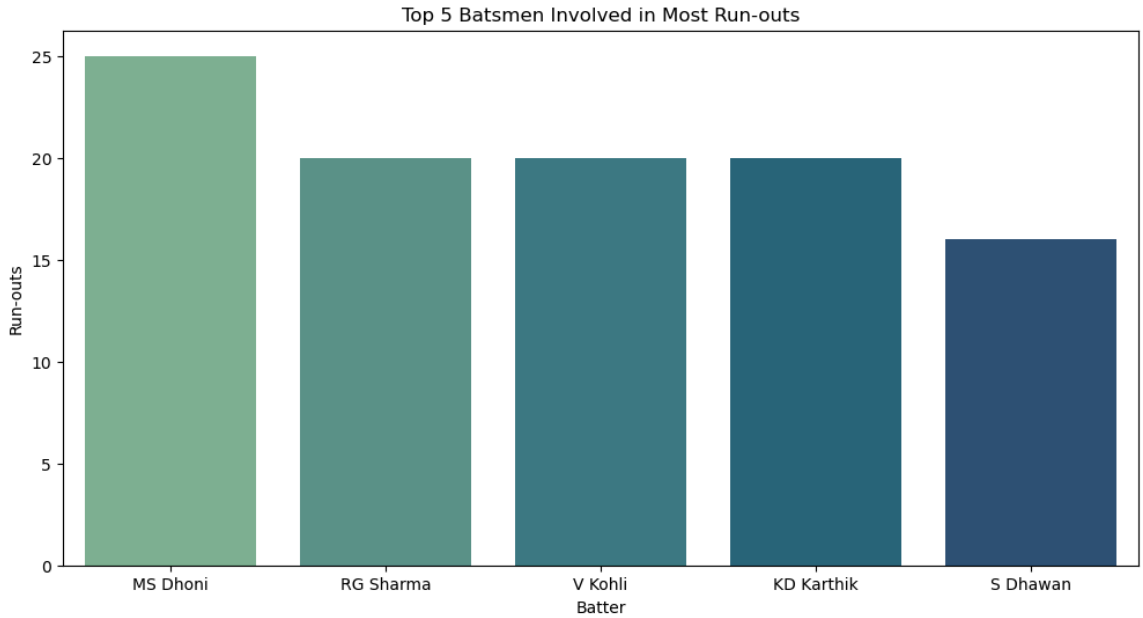
In [118]:

```
#5. Tabulate and visualize the batsmen (top 5) involved in the most run

run_outs = df[df["dismissal_kind"] == "run out"]["batter"].value_counts
print(run_outs)
plt.figure(figsize=(12,6))
sns.barplot(x=run_outs.index, y=run_outs.values, palette="crest")
plt.title("Top 5 Batsmen Involved in Most Run-outs")
plt.xlabel("Batter")
plt.ylabel("Run-outs")
plt
```

```
MS Dhoni      25
RG Sharma     20
V Kohli       20
KD Karthik    20
S Dhawan      16
Name: batter, dtype: int64
```

Out[118]:

```
<module 'matplotlib.pyplot' from 'C:\\ProgramData\\Anaconda3\\lib\\site-pa
ckages\\matplotlib\\pyplot.py'>
```



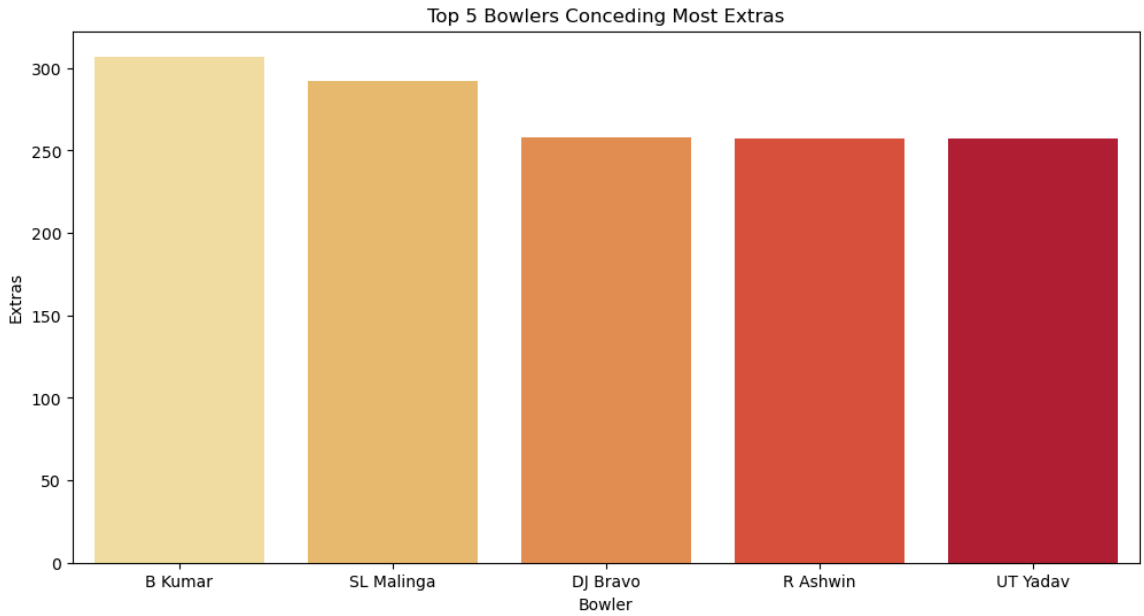
In [119]:

```
#6. Tabulate and visualize the top 5 bowlers who conceded the most extr

bowlers_extras = df.groupby("bowler")["extra_runs"].sum().nlargest(5)
print(bowlers_extras)
plt.figure(figsize=(12,6))
sns.barplot(x=bowlers_extras.index, y=bowlers_extras.values, palette="Y
plt.title("Top 5 Bowlers Conceding Most Extras")
plt.xlabel("Bowler")
plt.ylabel("Extras")
plt
```

```
bowler
B Kumar       307
SL Malinga    292
DJ Bravo      258
R Ashwin      257
UT Yadav      257
Name: extra_runs, dtype: int64
```

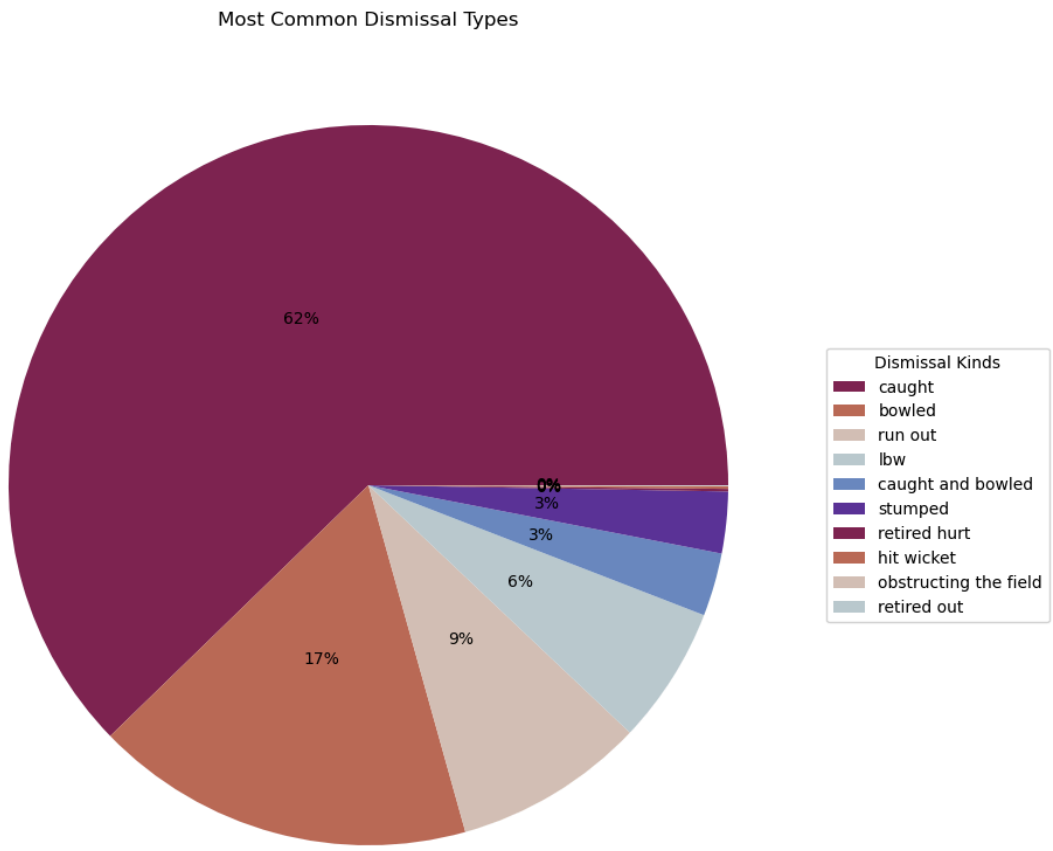
Out[119]:

```
<module 'matplotlib.pyplot' from 'C:\\ProgramData\\Anaconda3\\lib\\site-pa
ckages\\matplotlib\\pyplot.py'>
```



In [110]:

```
#7. Tabulate and visualize the most common dismissal type.

dismissal_kinds = df['dismissal_kind'].value_counts().reset_index()
dismissal_kinds.columns = ['dismissal_kind', 'count']
plt.figure(figsize=(10, 10))
palette_color = sns.color_palette('twilight_shifted_r')
plt.pie(dismissal_kinds['count'],colors=palette_color,textprops={'fonts

plt.legend(dismissal_kinds['dismissal_kind'], title="Dismissal Kinds",
plt.title('Most Common Dismissal Types')
plt.show()
```



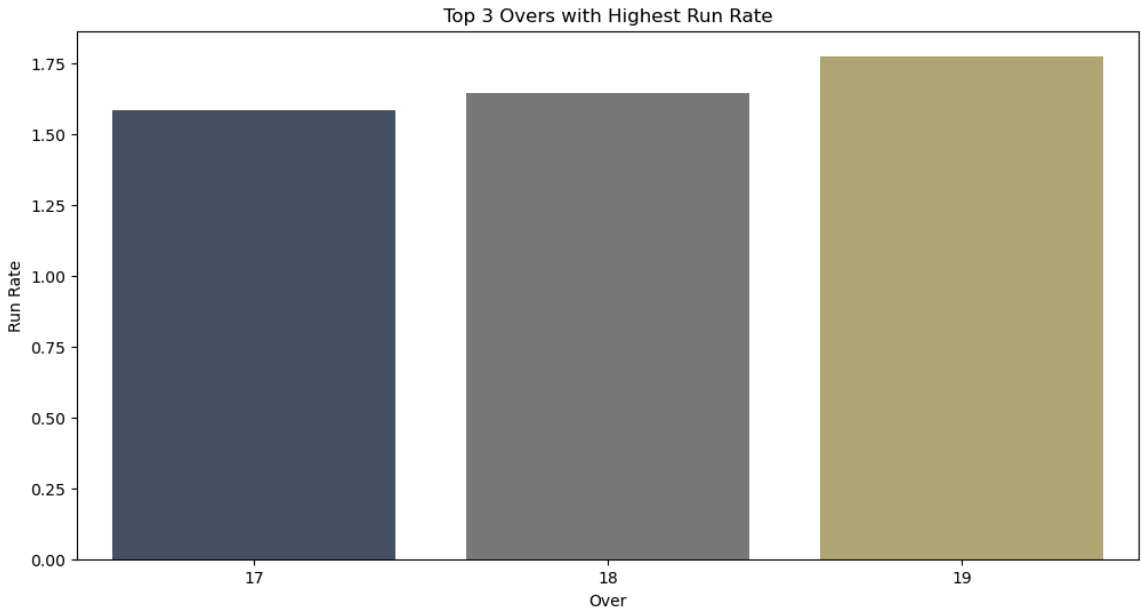
In [120]:

```
#8. Find and display which of the top 3 overs has the highest run rate

overs_runrate = df.groupby("over")["total_runs"].sum() / df["over"].val
top_overs = overs_runrate.nlargest(3)
print(top_overs)
plt.figure(figsize=(12,6))
sns.barplot(x=top_overs.index, y=top_overs.values, palette="cividis")
plt.title("Top 3 Overs with Highest Run Rate")
plt.xlabel("Over")
plt.ylabel("Run Rate")
plt

```

```
over
19    1.776855
18    1.646896
17    1.587839
dtype: float64
```

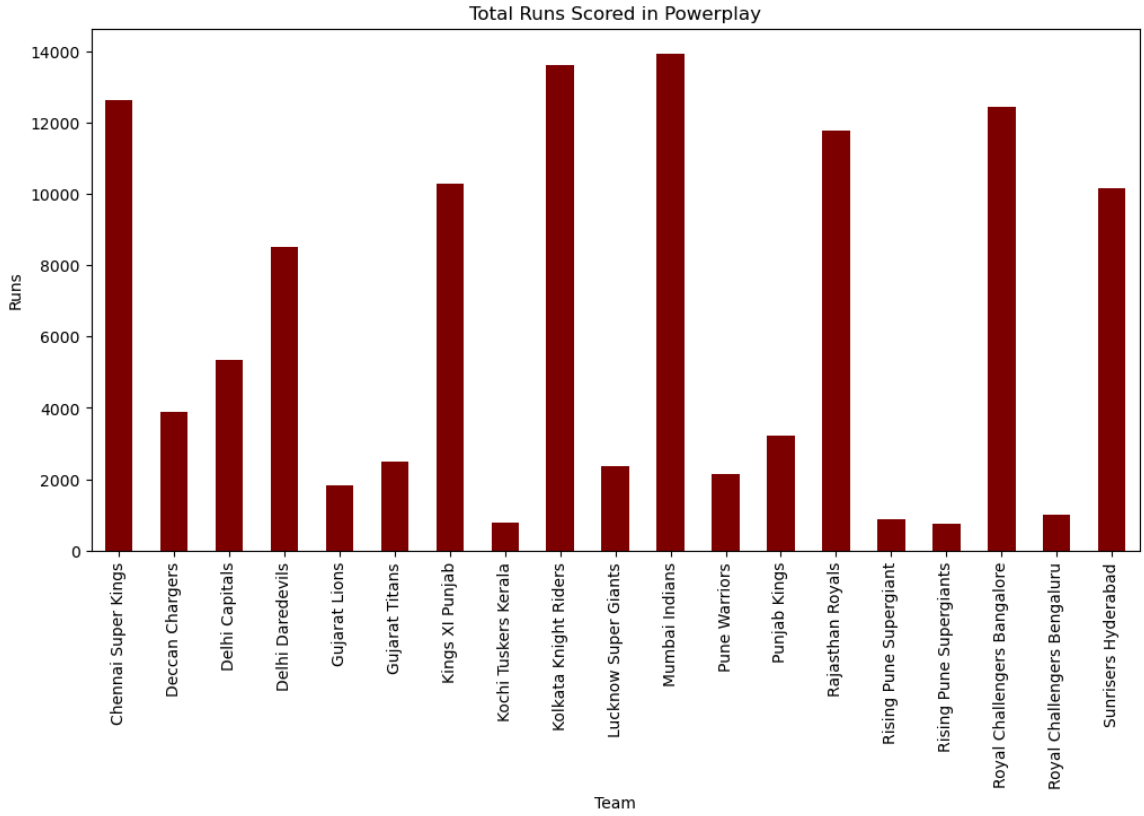
Out[120]:

```
<module 'matplotlib.pyplot' from 'C:\\ProgramData\\Anaconda3\\lib\\site-pa
ckages\\matplotlib\\pyplot.py'>
```



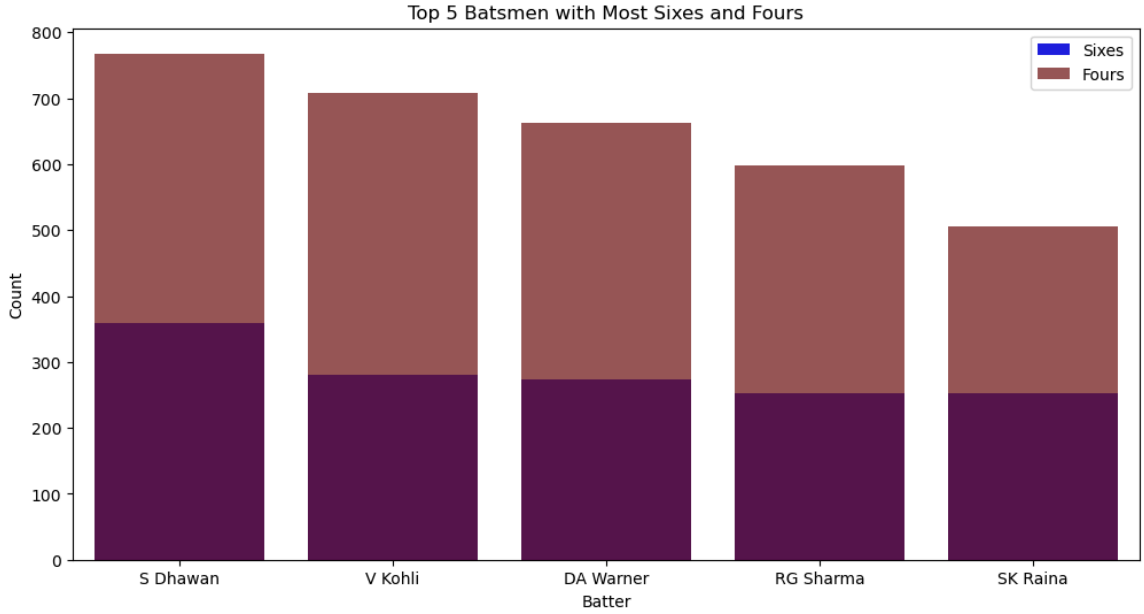
In [114]:

```
#9. Analyze and visualize powerplay performance (first 6 overs)

powerplay_runs = df[df["over"] <= 6].groupby("batting_team")["total_run
powerplay_runs.plot(kind='bar', figsize=(12,6), color='maroon', title='
plt.xlabel("Team")
plt.ylabel("Runs")
plt.show()
```



In [113]:

```
#10. Tabulate and visualize the top 5 batsmen with the most sixes and fo

sixes = df[df["batsman_runs"] == 6].groupby("batter")["batsman_runs"].co
fours = df[df["batsman_runs"] == 4].groupby("batter")["batsman_runs"].co
plt.figure(figsize=(12,6))
sns.barplot(x=sixes.index, y=sixes.values, color='blue', label='Sixes')
sns.barplot(x=fours.index, y=fours.values, color='maroon', label='Fours
plt.legend()
plt.title("Top 5 Batsmen with Most Sixes and Fours")
plt.xlabel("Batter")
plt.ylabel("Count")
plt.show()
```



In [ ]:

```
```